

Measuring Forces and Energy Through IoT Sensors

TECH204 Final Course Project

2025 May Session

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Introduction

This Project is a six-part experimental series designed to explore and apply the foundational physics concepts using IoT technology. Each part builds upon the last, resulting in a full-scale scientific investigation that incorporates real-time data, programming, and data analysis.

Throughout this project, I applied the scientific method to investigate motion, gravitational acceleration, energy conservation, rotational mechanics, and electromagnetic fields. These experiments not only reinforce theoretical physics knowledge but also developed my technical proficiency in using smart sensors and analyzing physical systems, key skills for a career in technology and Cybersecurity.

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Project Part 1

Inventory of Parts and Software

Ultrasonic Sensor HC-SR04

Describe the working principles of ultrasonic sensors in detail.

Answer: The Ultrasonic Sensor HC-SR04 is an electronic device with four pins: the Vcc pin, which is the first pin to the left, is the operating voltage pin; the Trig pin, is used to send the ultrasonic pulse; the Echo pin receives signal to measure the time of flight; and the GND pin the ground connection, completing the circuit. This sensor when activated, one transducer sends a burst of ultrasounds by converting electric pulses in sound waves, emitting a 40kHz sound wave from the Trig pin. The other transducer listens for the echo through the Echo pin. By calculating the time it takes for the sound to bounce off an object and return, the distance to the object can be determined using the speed of sound.

Rotary Encoder KY-040

Describe the working principles of rotary encoders in detail.

Answer: The Rotary Encoder Part KY-040 is an electro-mechanical device with five pins. The first pin to the left is GND, which connects to the ground of the board and completes the circuit. The Vcc pin supplies the power to the encoder. The CLK (Clock) pin sends pulses when you turn the knob and it helps determine the direction and speed of rotation. The DT (Data) pin works with the CLK pin. When the encoder turns, both the CLK and the DT generate square waves signals that are slightly offset in time. Inside the encoder is a small disk with metal contacts. As the disk spins, the contacts trigger CLK and DT in a specific pattern. If CLK goes high before DT, the encoder is being turned clockwise. If DT goes higher before CLK, it is being turned counterclockwise. The SW (Switch) pin is a button press. If you push the encoder knob down, it sends a signal through the SW pin. This device is an incremental rotary encoder, meaning it generates a continuous stream of electrical pulses in response to rotational movement, rather than tracking a position.

Hall Effect Sensor in ESP32

Describe the working principles of the Hall Effect Sensor in detail.

Answer: The Hall Effect Sensor in the ESP32 micro-controller is located under the metal lid you see on the top of the device. This sensor is designed to detect the presence of magnetic fields. Inside the sensor, a constant current flows through a thin conductive strip. When a magnetic field is applied perpendicular to this current, it interacts with the flow of electrons. This interaction causes the charged particles (electrons) to shift to one side of the strip, creating a voltage difference across the other side of the strip, which is the Hall Voltage. The sensor measures this voltage, and the greater the magnetic field strength, the higher the voltage output. The ESP32 reads this voltage and converts it into digital data to determine the presence, strength, and polarity of the magnetic field.

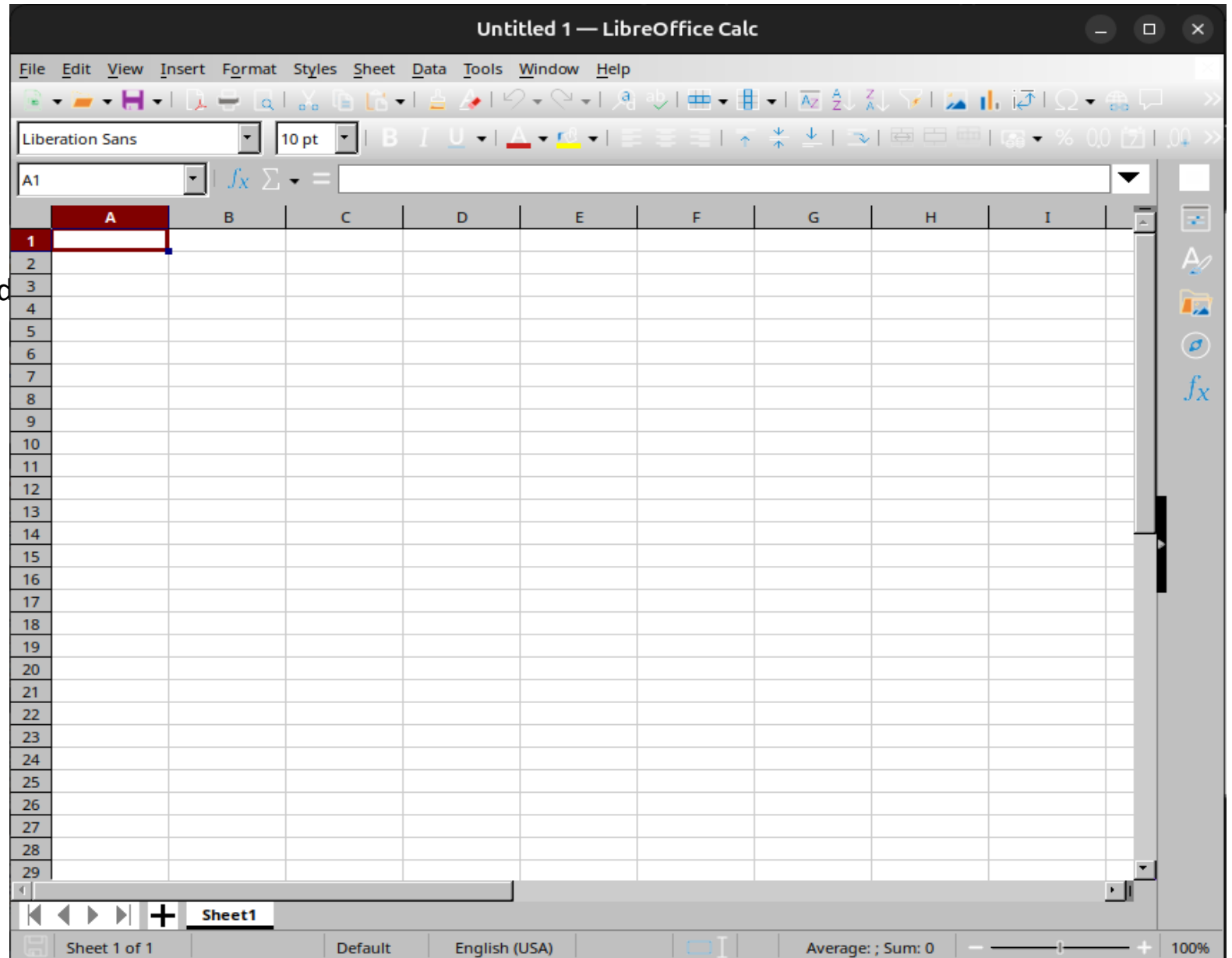
Screenshots

What tool will you used to take screenshots? How does it work?

Answer: On Ubuntu, I use the built-in Screenshot tool to take my screen shots. This tool allows you to capture the entire screen, a selected area, or just the active window using simple keyboard shortcuts. For example, you can press Prtsc to capture the full screen, while shift + Prtsc lets you select a specific area. The tool works by capturing the pixel data currently displayed and saving it as an image file that then can be inserted into documents or presentations.

Required Software

Screenshot of Microsoft Excel installed and running on your computer



Conclusion

This section introduces the foundational elements of the course project, focusing on the equipment and software used to analyze systems. It includes an inventory of sensors and tools necessary to replicate scientific experiments related to motion, energy, and force.

The objective of this part was to:

- Identify the materials and sensors required for the remaining experiments
- Understand the function of devices as ultrasonic sensors, rotary encoders, and Hall effect sensors
- Install and prepare software needed to analyze and process sensor data

By preparing a clear inventory and familiarizing myself with tools, I laid the groundwork for reproducible, scientific, and structured approach to each experiment in this series.

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Project Part 2

Precision of the Ultrasonic Sensor

Data Collection

Trial	Ruler Distance (cm)	Total Roundtrip Distance (m)	Time from Serial Monitor (microseconds)	Roundtrip time (s)	Velocity = distance/time (m/s)
1	8	0.16m	492	0.000492s	325.20 m/s
2	10	0.20m	586	0.000586s	341.30 m/s
3	15	0.30m	925	0.000925s	324.32 m/s
4	20	0.40m	1187	0.001187s	336.98 m/s
5	30	0.60m	1724	0.001724s	348.03 m/s

Data Analysis

- Average velocity from table

$$v_{avg} = \frac{v_1 + v_2 + v_3 + v_4 + v_5}{5}$$

Answer: 335.17 m/s
units

- Percent difference where $v_{sound} = 343 \text{ m/s}$

$$\text{Percent difference} = \frac{|v_{avg} - v_{sound}|}{v_{sound}} \times 100$$

Answer: 2.28 %
units

Questions and Answers

Briefly describe and explain the results.

Answer: The experimental results was very close to the theoretical speed of sound of 343 m/s. The percent difference was only 2.28%, which is under the 5% threshold. This 2% difference shows that the Ultrasonic Sensor performed reliably. Minor errors could be caused by timing precision, environmental conditions, or slight misalignment during measurement. If the object shape were different, such as curved or textured, it could effect how the sound reflects, slightly altering the results.

What is the dependence of temperature, humidity, and atmospheric pressure on the speed of sound? How does it affect the results?

Answer: The speed of sound depends on temperature and humidity. Theoretically, it is calculated using:

Although pressure and density both increase or decrease together, their effects cancel out, so atmospheric pressure has little to no effect on sound when temperature is constant.

$$c = \sqrt{\frac{\gamma P}{\rho}}$$

At 0°C, the speed of sound in air is about 1087.42 ft/s (331.45 m/s). As temperature increases, so does sound speed at 20°C, its around 343 m/s. Humidity also increases sound speed because water vapor is less dense than dry air.

These changes matter in sensor reading and experiments, which is why calculated velocity might slightly differ from the standard 343 m/s.

Conclusion

In this experiment, an ultrasonic sensor was used to calculate distance by measuring the round-trip of a sound pulse bouncing off an object. The goal was to determine how precise the sensor was by comparing calculated velocities to the known speed of sound (343 m/s). This experiment helped assess the reliability of sensor data and emphasized how factors like temperature and humidity affect accuracy.

The steps included:

- Measuring the round-trip time of sound at five known distances
- Calculating individual velocities for each trial
- Finding the average velocity
- Computing the percent difference from the known speed of sound

This reinforced the importance of accurate data collection and environmental awareness when working with real-world sensors.

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Project Part 3

Gravitational Acceleration of a Free-Falling Object

Excel Table (Screenshot)

Include all values in the Excel table for a single trial showing the fall distance in centimeters and meters as a function of time

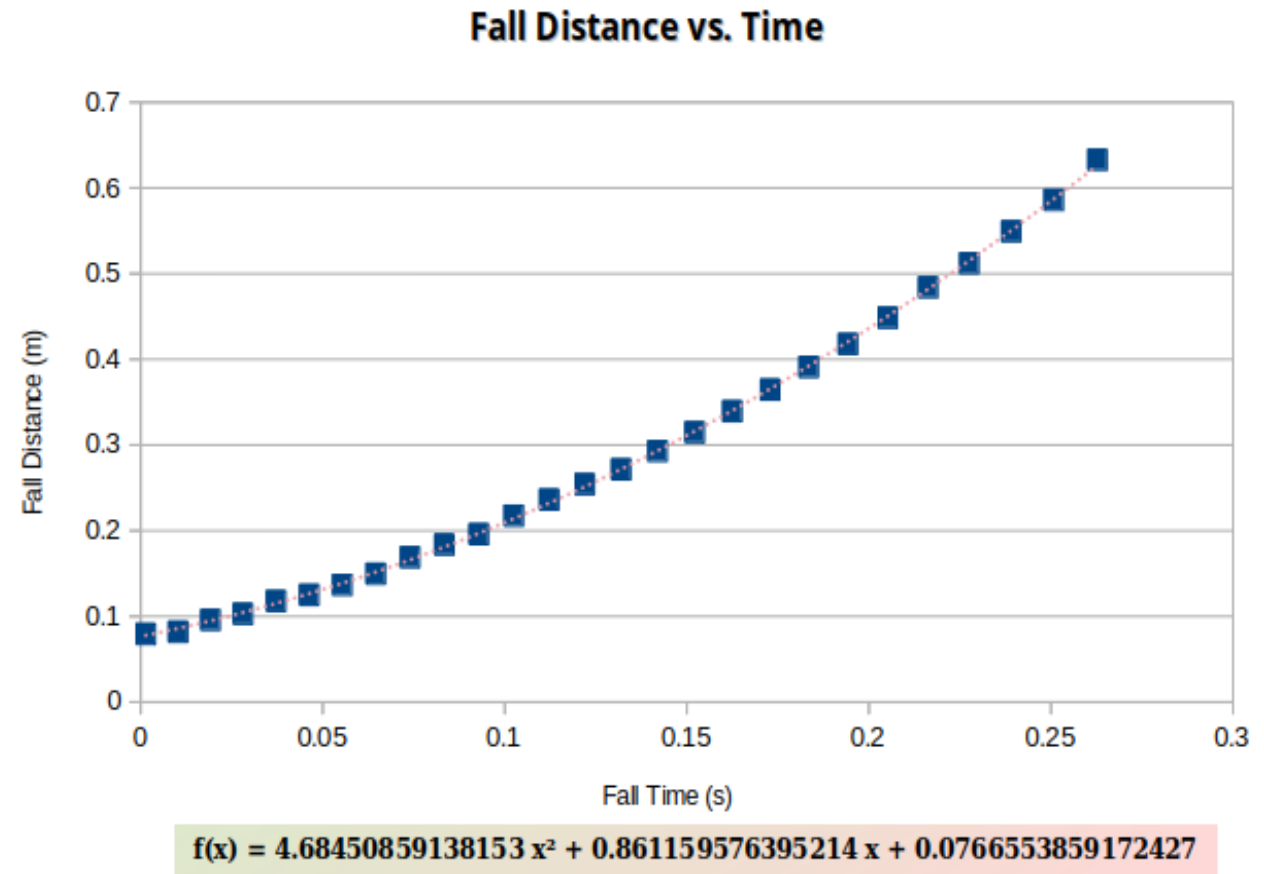
TECH204 Module 3 Project		
Name TECH204 Team 1		
Speed of sound: 343.00 m/s		
Initial distance to ground: 69.29		
TIME (s)	Distance (cm)	Distance (m)
0.0014	7.94	0.0794
0.0103	8.2	0.082
0.0192	9.57	0.0957
0.0281	10.29	0.1029
0.0372	11.78	0.1178
0.0463	12.5	0.125
0.0554	13.67	0.1367
0.0646	14.97	0.1497
0.074	16.89	0.1689
0.0834	18.38	0.1838
0.0929	19.59	0.1959
0.1025	21.75	0.2175
0.1122	23.62	0.2362
0.122	25.42	0.2542
0.132	27.15	0.2715
0.142	29.29	0.2929
0.1522	31.49	0.3149
0.1625	33.94	0.3394
0.1729	36.46	0.3646
0.1835	39.1	0.391
0.1943	41.79	0.4179
0.2052	44.85	0.4485
0.2163	48.4	0.484
0.2276	51.21	0.5121
0.2391	54.93	0.5493
0.2508	58.6	0.586
0.2628	63.28	0.6328

Excel Graph (Screenshot)

Include Excel graph for a single trial showing the fall distance as a function of time.

Confirm that most data points fall along the curve fitting.

The x^2 coefficient should be between 2.5 and 7.5 such that it yields an acceleration value between 5 m/s^2 and 15 m/s^2



Data Collection

Trial	x ² coefficient value from curve fitting	Acceleration (m/s ²) = 2* (x ² coefficient)
1	4.6845	9.369 m/s ²
2	5.0203	10.0406 m/s ²
3	4.6635	9.327 m/s ²
4	4.6161	9.23398 m/s ²
5	4.7269	9.4538 m/s ²

Data Analysis

- Average acceleration from table

$$a_{avg} = \frac{a_1 + a_2 + a_3 + a_4 + a_5}{5}$$

Answer: 9.485 m/s²
units

- Percent difference with $g = 9.8 \text{ m/s}^2$

$$\text{Percent difference} = \frac{|a_{avg} - g|}{g} \times 100$$

Answer: 3.22 %
units

Questions and Answers

Briefly describe and explain the results.

Answer: The data we collected showed a curved pattern between how far the object fell and how much time passed. This curve is what we would expect from something falling under gravity. We used the X squared value from our Excel graphs to figure out the acceleration. All five trials gave us results really close to Earth's actual gravity, with our average being 9.8 m/s. That tells us our setup and method were pretty accurate.

How much variation is there from trial to trial? What does this indicate about the uncertainty of the result?

Answer: The calculated acceleration values varied slightly across the five trials, with a highest percent difference of about 3.22%. This small variation indicates that the set up was consistent with low uncertainty.

Conclusion

This experiment used an ultrasonic sensor to track the vertical motion of a free-falling object and calculate the acceleration due to gravity (g). Starting from rest, the object's position was measured over time and modeled using the kinematic equation:

$$y(t) = \frac{1}{2}gt^2 + y_0$$

The goal was to extract the value of g from the quadratic fit of the position-time data, ideally arriving close to the theoretical value of 9.80 m/s^2 .

Through this experiment, I gained insight into:

- How motion can be modeled mathematically using kinematic equations
- How sensor data can be used to graph real-time free-fall motion
- How to fit a curve and extract physical constants (like g) from real-world data

This part also laid the foundation for part 4, where energy conservation will be explored using the same motion data.

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Project Part 4

Conservation of Energy of a Free-Falling Object

Excel Table (Screenshot)

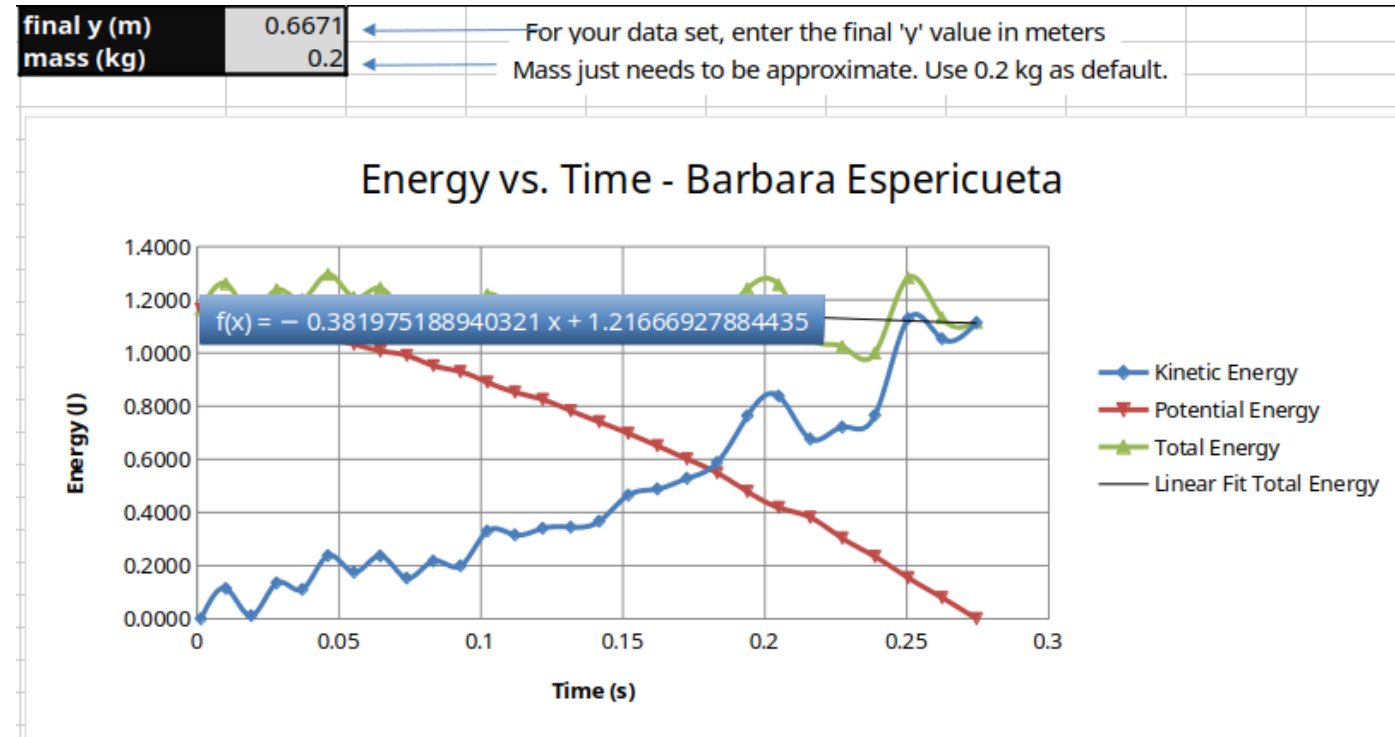
Include all values in the Excel table for a single trial showing the fall distance in centimeters and meters as a function of time with the values generated for kinetic energy K, potential energy U, and total mechanical energy E.

t(s)	y(cm)	y (m)	h (m)	v(m/s)	K(J)	U(J)	E(J)
0.0014	7.31	0.0731	0.59	0.00	0.0000	1.1642	1.1642
0.0102	8.25	0.0825	0.58	1.07	0.1141	1.1458	1.2599
0.0191	8.56	0.0856	0.58	0.35	0.0121	1.1397	1.1519
0.0281	10.41	0.1041	0.56	1.16	0.1348	1.1035	1.2383
0.0371	11.08	0.1108	0.56	1.05	0.1107	1.0903	1.2010
0.0462	12.74	0.1274	0.54	1.54	0.2379	1.0578	1.2957
0.0553	14.01	0.1401	0.53	1.32	0.1752	1.0329	1.2081
0.0646	15.31	0.1531	0.51	1.54	0.2366	1.0074	1.2440
0.0739	16.17	0.1617	0.51	1.24	0.1533	0.9906	1.1439
0.0833	18.14	0.1814	0.49	1.48	0.2176	0.9520	1.1695
0.0928	19.29	0.1929	0.47	1.41	0.1992	0.9294	1.1286
0.1023	21.33	0.2133	0.45	1.82	0.3301	0.8894	1.2196
0.112	23.24	0.2324	0.43	1.78	0.3158	0.8520	1.1678
0.1218	24.64	0.2464	0.42	1.84	0.3403	0.8246	1.1649
0.1317	26.79	0.2679	0.40	1.86	0.3449	0.7824	1.1273
0.1417	28.93	0.2893	0.38	1.92	0.3670	0.7405	1.1075
0.1519	31.13	0.3113	0.36	2.16	0.4649	0.6974	1.1623
0.1622	33.53	0.3353	0.33	2.21	0.4883	0.6503	1.1387
0.1726	36.03	0.3603	0.31	2.30	0.5280	0.6013	1.1293
0.1832	38.72	0.3872	0.28	2.42	0.5880	0.5486	1.1366
0.1939	42.29	0.4229	0.24	2.76	0.7636	0.4786	1.2423
0.2049	45.38	0.4538	0.21	2.89	0.8380	0.4181	1.2560
0.216	47.25	0.4725	0.19	2.60	0.6763	0.3814	1.0577
0.2273	51.26	0.5126	0.15	2.69	0.7213	0.3028	1.0241
0.2388	54.76	0.5476	0.12	2.77	0.7656	0.2342	0.9998
0.2505	58.84	0.5884	0.08	3.36	1.1286	0.1543	1.2828
0.2624	62.65	0.6265	0.04	3.25	1.0530	0.0796	1.1326
0.2746	66.71	0.6671	0.00	3.34	1.1142	0.0000	1.1142

Excel Graph (Screenshot)

Include Excel graph for a single trial showing the plots of the kinetic energy K, potential energy U, and total mechanical energy E as a function of time.

Must include your name in the title of the graph.



Data Analysis

Theoretical value of final velocity

$$h = y_{final} - y_{initial} = \frac{0.594}{\text{units}} \quad \text{m}$$

$$v_f = \sqrt{2gh} = \frac{3.41}{\text{units}} \quad \text{m/s}$$

Final velocity from experimental data (or largest velocity)

$$v_{experimental} = \frac{3.34}{\text{units}} \quad \text{m/s}$$

Percent difference

$$\text{Percent difference} = \frac{|v_f - v_{experimental}|}{v_f} \times 100$$

$$\text{Answer: } \frac{2.05}{\text{units}} \quad \%$$

Questions and Answers

Discuss the key characteristics of the plot. Consider the points when potential energy U is maximum, U is minimum, kinetic energy K is maximum, K is minimum and when U and K are the same value. What is the significance of these points?

Answer: At the beginning of the fall, potential energy (U) is at its maximum because the object is at its highest point, and kinetic energy (K) is at its minimum because it hasn't started moving. As the object falls, U decreases while K increases. The point where U and K are equal represents the midpoint of the fall, where energy is equally distributed between height and speed. At the bottom of the fall, K is maximum and U is minimum, nearly all potential energy has been converted into kinetic energy. The significance lies in transformation and conservation of mechanical energy throughout the free fall.

Questions and Answers Continued

What is the trend of the best fit line for the total energy E in your data? If the data is accurate, the total mechanical energy should decrease slightly. Why is that?

Answer: The best fit line for the total mechanical energy (E) shows a slight negative slope, indicating a small decrease in total energy over time. While mechanical energy is theoretically conserved, this small decline is expected due to experimental factors such as air resistance, sensor timing errors, and the use of average velocity instead of instantaneous velocity. These small inaccuracies result in a slight loss of calculated energy, even though the overall trend still supports conservation of energy.

Conclusion

In this experiment, I analyzed the energy transforming of a free-falling object to demonstrate the conservation of mechanical energy. Using previously collected motion data, I calculated both gravitational potential energy ($u=mgy$) and kinetic energy ($k = 1/2mv^2$) at various points during the fall.

The goal was to verify that the total mechanical energy ($E = K + U$) remained consistent throughout the motion, confirming the principle that, in the absence of friction or air resistance, energy is conserved.

Through this experiment, I:

- Calculated changing potential and Kinetic energy as the object fell
- Observed how potential energy converted into kinetic energy
- Verified that the total energy stayed approximately consistent despite individual values changing

This experiment deepened my understanding of how motion and energy are connected and how real-world data can reveal fundamental physics laws.

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Project Part 5

Relationship between Linear and Rotational Motion

Data Collection

- Object diameter: $d =$
units
- Object radius: $r =$
units
- Number of pulses for one revolution: $x =$
- Resolution = $\left(\frac{1}{x} \frac{\text{Revolution}}{\text{pulses}} \right) \cdot \left(\frac{2 \cdot \pi \text{ radians}}{1 \text{ Revolution}} \right) =$
units

Data Analysis

Trial	Number of pulses N	Trial	Number of pulses N	Encoder distance r Resolution N	Measured distance with ruler	Percent difference	Measured distance with ruler	Percent difference
		1						
		2						
		3						
		4						
		5						
1	14	0.077					0.08	3.75
2	17	0.093					0.10	7.00
3	21	0.115					0.12	4.17
4	27	0.148					0.14	5.71
5	30	0.165					0.16	3.13

Questions and Answers

Briefly describe and explain the results.

Answer: I calculated the encoder distance for each trial using the arc length formula. By plugging in the same radius and resolution values for each trial, I was able to consistently calculate the expected linear distance from the pulse count. The measured distance taken with a ruler were close to the calculated distance, and the percent difference were mostly low. This shows the math and coder setup worked perfectly. Any small errors were likely due to measurement or rolling inconsistencies.

What are some applications of this measuring technique? Could this measuring system be used to measure surfaces that are not flat?

Answer: A rotary encoder is a motion-sensing device used to detect position, speed, and direction of rotating parts. It's commonly used in robotics, elevators, industrial automation, and CNC machines to measure angular rotation and convert it into precise linear motion. While this method is most accurate on flat surfaces, uneven surfaces may require additional sensors or compensation for slippage. (Source: Dynapar, n.d.)

Conclusion

This experiment introduced circular motion by using a rotary encoder to convert rotational movement into linear displacement. By rolling a circular object attached to the encoder across a surface, the number of electrical pulses generated was used to calculate the angle of rotation and, using the object's radius, the total linear distance traveled.

The experiment applies the following relationships:

- Angular Displacement (θ) = (Number of Pulses / Pulses per Revolution) $\times 2\pi$
- Linear Displacement (d) = Radius $\times \theta$

This part of the project reinforced how:

- Real-world sensors can measure angular rotation
- Geometry and trigonometry apply in circular motion
- Rotary encoders are used in robotics, automation, and IoT devices to track position

Understanding how rotary motion maps to linear motion bridges the gap between simple physics and real-world engineering systems.

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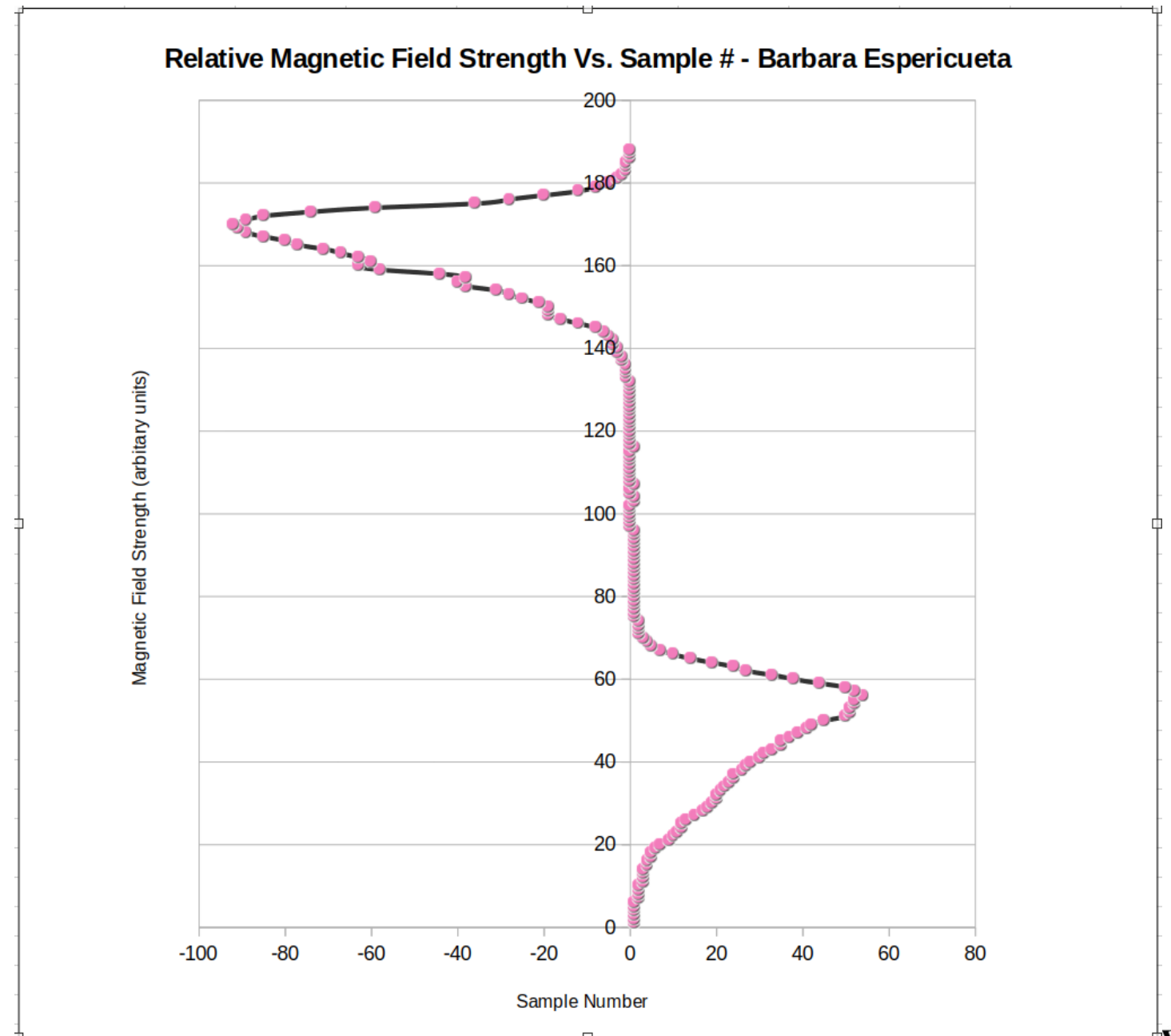
Project Part 6

Observing the Hall Effect

Excel Graph (Screenshot)

Include Excel graph for a single trial of data plotted as a function of time.

Must include your name in the title of the graph.



Questions and Answers

Briefly describe and explain the results.

Answer: The data shows how the magnet field strength increased and decreased as the magnet was moved near the Hall sensor. The sharp flip in the graph represents the moment the magnet's poles were reversed, either from north to south or vice versa. This caused the electrons in the Hall sensor to shift in the opposite direction, changing the polarity of the voltage. The closer the magnet got, the sharper the field, and the more noticeable the shift in the data. This flip and slope clearly reflect the positive and negative pull of the magnet as it passed over the sensor.

Describe how the Hall sensor works. What are some applications for the Hall effect sensor?

Answer: The Hall effect sensor detects the presence and strength of a magnetic field by measuring the voltage generated perpendicular to both the electric current and the magnetic field in a conductor. When a magnetic field is present, it detects the moving charge, creating a measurable voltage. This voltage is directly related to the strength of the magnetic field. Hall effect sensors are used in many everyday technologies. In cars, they are used in ABS systems and to measure wheel speed. They are also used in simple devices like laptops, where the sensor detects when the lid is closed by sensing a magnet, allowing the system to go to sleep or wake up.

Conclusion

In this final experiment, I used an ESP32 micro-controller and built-in Hall sensor to detect magnetic fields and demonstrate the Hall Effect, a phenomenon where voltage is generated when a magnetic field interacts with a current-carrying conductor.

By bringing a magnet close to the ESP32's Hall sensor, I observed voltage values that increase or become negative depending on the polarity and distance of the magnetic field.

The goal of this experiment were to:

- Measure how magnetic field strength affects sensor readings
- Identify the relationship between magnetic polarity and voltage
- Create a plot showing real data points as the magnetic field changes

This experiment connected abstract electromagnets theory with a real, observation system. It also demonstrated how IoT devices can detect and respond to invisible physical forces like magnetic fields, essential in everything from digital compasses to brushless motors.

Challenges/Lessons Learned

One of the biggest challenges I faced during this project was managing all the different physics formulas and keeping them organized. With so many numbers and equations being used across each part, it was easy to get overwhelmed or accidentally mix things up.

To overcome this, I dedicated a notebook just for physics class and wrote down every formula, concept, and project step in detail. This helped me stay organized and build confidence as I worked through each experiment.

The biggest lesson I learned is that physics might seem intimidating at first, but once you get hands-on and break it down step by step, it becomes incredibly fun and rewarding. I gained not only knowledge, but also trust in my ability to solve complex problems through persistence and curiosity.

Career Skills

Throughout this course project, I developed valuable technical and analytical skills that will directly benefit my career. These include data analysis, sensor integration, interpreting real-time feedback, and applying mathematical models to solve real-world problems.

This project also strengthened my critical thinking, time management, and communication skills — all essential in technology, engineering, and cybersecurity fields. By working with IoT hardware and the scientific method, I built competencies in experimentation, system design, and troubleshooting, which will help me qualify for hands-on roles in STEM and prepare for more advanced certifications and career opportunities.

Conclusion

This course project gave me the opportunity to explore the physical world through hand-on experimentation and real sensor data. Each part of the project built on the last, allowing me to apply the scientific method, practice critical thinking, and strengthen my understanding of fundamental physics concepts.

By connecting theory with technology, I developed skills that not only support academic success but also lay the foundation for my future in the tech industry. From measuring motion and energy to observing electromagnet effects, I learned how powerful and relevant physics is in the modern innovation.

Completing this project reminded me that even the most challenging concepts can become clear and rewarding with patience, organization, and curiosity.

References

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